

Recommendation Systems for Personalized Content Discovery

Participation Format

This is a team problem statement with a maximum of 02 participants per team.

Teams will work with the Netflix Prize Dataset to develop a personalized recommendation system capable of learning user preferences and generating relevant content recommendations.

1. Motivation

Every day, billions of users interact with recommendation systems that influence the movies they watch, the songs they listen to, the products they purchase, and the content they consume online.

Recommendation systems have become one of the most impactful applications of Artificial Intelligence and Machine Learning, powering personalization across platforms such as Netflix, Amazon, Spotify, YouTube, LinkedIn, and Instagram.

The ability to accurately understand user preferences and deliver relevant content directly influences user engagement, retention, and overall user experience. As a result, recommendation systems have evolved into a critical component of modern digital products.

One of the most influential milestones in the history of recommender systems was the Netflix Prize, a global machine learning competition launched to improve movie recommendation accuracy. The dataset released through this initiative remains one of the most widely studied datasets in recommendation system research and industry.

In this challenge, participants will step into the role of Machine Learning Engineers and Data Scientists tasked with building an intelligent recommendation engine capable of predicting user preferences and generating personalized content recommendations.

Dataset Overview

Dataset Link

<https://www.kaggle.com/datasets/netflix-inc/netflix-prize-data>

Participants will work with the Netflix Prize Dataset, one of the most influential datasets in the history of recommender systems.

Originally released as part of the Netflix Prize competition, the dataset was created to improve Netflix's movie recommendation algorithm and has since become a benchmark dataset for recommendation system research.

The dataset contains:

- 100,480,507 movie ratings
- 480,189 users
- 17,770 movies
- Ratings on a 1–5 star scale
- Rating timestamps for each interaction
- Movie metadata including movie IDs, titles, and release years

Each rating record consists of:

- User ID
- Movie ID
- Rating
- Date of Rating

The dataset is highly sparse, meaning that each user has rated only a small fraction of the available movies. This characteristic makes it particularly suitable for studying real-world recommendation challenges such as collaborative filtering, latent factor modeling, recommendation ranking, and cold-start problems.

Participants are encouraged to use appropriate subsets of the data if computational constraints prevent training on the full dataset.

2. Problem Statement

You are part of the machine learning team responsible for improving content discovery on a large-scale streaming platform.

Using historical user-item interaction data from the Netflix Prize Dataset, your task is to design and develop a recommendation system capable of delivering personalized content recommendations to users.

The recommendation engine should be capable of:

- Learning user preferences from historical interactions
- Predicting ratings for unseen content
- Generating personalized recommendations
- Identifying similarities between users and content items
- Improving content discovery through relevant recommendations

Participants may explore a variety of recommendation approaches, including but not limited to:

- User-Based Collaborative Filtering
- Item-Based Collaborative Filtering
- Matrix Factorization
- Singular Value Decomposition (SVD)
- Alternating Least Squares (ALS)
- Neural Collaborative Filtering
- Hybrid Recommendation Systems
- Ensemble Methods

The objective is not merely to predict ratings but to build a recommendation system that effectively captures user preferences and delivers meaningful recommendations.

3. Mandatory and Optional Tasks

Mandatory Tasks

A. Exploratory Data Analysis

Perform a detailed analysis of the dataset and identify:

- User activity patterns
- Content popularity trends
- Rating distributions
- Data sparsity characteristics

- Interesting observations and insights

Participants should clearly communicate the business and technical implications of their findings.

B. Recommendation Model Development

Develop at least one recommendation model capable of:

- Learning user preferences
- Predicting unseen ratings
- Generating personalized recommendations

Clearly explain the methodology used and justify its suitability for the problem.

C. Model Comparison

Implement and compare at least two recommendation approaches.

Possible comparisons may include:

- User-Based vs Item-Based Collaborative Filtering
- Matrix Factorization vs Collaborative Filtering
- Traditional Methods vs Deep Learning Approaches

Participants should discuss:

- Recommendation quality
 - Training complexity
 - Computational efficiency
 - Practical usability
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D. Recommendation Generation

Generate Top-K recommendations for users and analyze the recommendations produced.

Participants should demonstrate:

- Sample recommendations
- Recommendation quality

- Success cases
 - Failure cases
 - Key observations
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E. Evaluation

Participants must evaluate their recommendation system using the following **mandatory metrics**:

Mandatory Metrics

- **RMSE (Root Mean Squared Error)** – for measuring rating prediction accuracy.
- **MAP@10 (Mean Average Precision @ 10)** – for measuring recommendation ranking quality.

For the purpose of calculating **MAP@10**, a movie should be considered **relevant if its actual user rating is greater than or equal to 3.5**.

Participants must clearly describe:

- Their train-test split methodology
- The relevance definition used
- The procedure followed to generate Top-10 recommendations
- The methodology used to compute MAP@10

Additional Metrics (Optional)

Participants may additionally report:

- MAE
- Precision@K
- Recall@K
- NDCG
- Hit Rate
- Coverage
- Diversity Metrics

Participants must justify their evaluation strategy and discuss the trade-offs between rating prediction accuracy and recommendation ranking performance.

Optional Tasks

A. Explainable Recommendations

Provide explanations for recommendations generated by the model.

Example:

Users who enjoyed Movie A and Movie B were also likely to enjoy Movie C.

B. Cold Start Strategy

Discuss approaches for handling:

- New users
- New content items
- Sparse interaction histories

Implementation is optional.

C. Interactive Dashboard

Build an interface that allows users to:

- View recommendations
- Explore similar content
- Analyze recommendation scores

This task is optional and will not affect eligibility for evaluation.

D. Hybrid Recommendation System

Incorporate additional metadata or external information to improve recommendation quality.

E. Deployment

Deploy the recommendation system as:

- A Web Application
- A Dashboard

- An API Service

Deployment is optional.

4. Deliverables

Deliverable 1: Technical Report

Format: PDF

The report should include:

- Problem Understanding
- Exploratory Data Analysis
- Methodology
- Model Design
- Evaluation Metrics
- Experimental Results
- Recommendation Examples
- Key Insights
- Future Improvements

Maximum Length: 10 Pages

Deliverable 2: Source Code & GitHub Repository

The submission must include a **GitHub repository** containing:

- Data Processing Pipeline
- Model Training Pipeline
- Evaluation Scripts
- Recommendation Generation Module
- Documentation
- Instructions to reproduce results

The repository should be well-structured, documented, and reproducible.

Deliverable 3: Presentation

Format: PDF

Maximum Length: 8 Slides

The presentation should cover:

- Problem Overview
 - Approach
 - Key Insights
 - Experimental Results
 - Recommendation Examples
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5. Guidelines

1. Focus on Recommendation Quality

The objective is not simply to build a predictive model but to generate useful and relevant recommendations.

2. Justify Your Decisions

Clearly explain:

- Model choices
 - Evaluation metrics
 - Feature engineering decisions
 - Recommendation strategy
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3. Balance Performance and Simplicity

A simpler model with strong insights and thoughtful analysis is preferable to an unnecessarily complex solution.

4. Encourage Experimentation

Participants are encouraged to compare multiple approaches and explore creative techniques for improving recommendation quality.

5. Think Like a Machine Learning Engineer

Strong submissions will demonstrate not only model-building skills but also an understanding of user behavior, recommendation quality, ranking performance, and practical deployment considerations.

6. Prioritize Reproducibility

Your work should be easy to understand, reproduce, and extend. Proper documentation and repository organization are strongly encouraged.

Evaluation Criteria

Criterion	Weightage
Data Understanding & EDA	15%
Recommendation Model Development	30%
RMSE & MAP@10 Performance	20%
Recommendation Quality & Insights	20%

Innovation & Creativity	10%
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Presentation & Documentation	5%
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